

Workshop 10

# **Automatic Speech Recognition**

Can a computer understand  
your voice?

# Task 1

- Download the sample audio.
- Try the following code.

```
import IPython.display as ipd  
ipd.Audio('sample.wav')
```

# ASR Pipeline

- Audio → Features → Acoustic Model → Decoder → Text
- Audio: The raw waveform
- Model: Maps the sound into letters
- Decoder: Turn probabilities into text

# Task 2

- Study the provided Jupyter notebook.
- Update the dataset to `test[:80%]`
  - There will be 2096 instances.
- Compare the performance of wav2vec and whisper ASR models using WER.
- Submission:
  - Python file that compares the results of the two models.

# Task 3

- Use an ASR model of your choice.
- Record the following sentences.
  - "I just came back from the bank"
  - "We love to play tennis"
  - "I can't tell whether the weather will affect where we're going"
  - "Dr. Shrestha emailed vmes@au.edu on December 3rd, 2025 at 11:59 p.m."
  - "The API uses HTTPS, JSON, and OAuth 2.0"
  - "The transformer's self-attention mechanism mitigates vanishing gradients during backpropagation"
  - I was born in—uh—no, actually I grew up in—wait—Chiang Mai.
  - I'll deploy the model after lunch, **เดี๋ยวเจอกันนะ**
- Run and evaluate the chosen ASR model with your recordings.

"See you in a bit"  
in your language

# Key Takeaways

- ASR  $\neq$  Text Classification
- Audio is messy.
- The pretrained models are powerful, but **not perfect**.
- Error analysis is crucial.
  - Matters more than the accuracy.